A Literature Review of Prosopagnosia

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Next time you run into someone on the street and they greet you with your full name while referencing memories that you two have shared, don't fret that you don't recognize them — you may finally have something in common with Brad Pitt and Jane Goodall: prosopagnosia! All jokes aside, although forgetting a face here and there is normal and is nothing to be concerned about, if you find yourself constantly unable to recognize faces that others say you should know, then perhaps this article can help you with familiarizing yourself with prosopagnosia.

An introduction to prosopagnosia

The ability to see someone and instantly recognize them is something that most people can naturally accomplish without much thought, but its relevance in daily functioning is often overlooked. For some, identifying and perceiving faces is a constant struggle, leading to an onslaught of challenges. Prosopagnosia, commonly known as face blindness, is the inability to recognize familiar faces. The neurodevelopmental disorder was first traced back to German neurologist Joachim Bodamer in 1947 where he describes prosopagnosia as agnosia for the recognition of faces and expressions. The face recognition disorder can be either the result of genetics in the absence of brain damage — developmental prosopagnosia (DP) — or acquired brain lesions or damage — acquired prosopagnosia.

DP is rare, affecting ~2% of the population, and is also a heterogeneous disorder, making prosopagnosia research and documentation limited due to the diverse character of the disorder (Susilo & Duchaine, 2013). As reported by Dalrymple et al. (2012), the possible causes for DP include genetics, malfunctioning of innate face processing mechanisms, experiential factors, and neurological factors. And, these factors are not mutually exclusive. DP tends to run in families, and recent self-report findings offer that DP could be an autosomal dominant disorder (Susilo & Duchaine, 2013). The first case of DP was documented by McConachie in 1976, describing AB, a 12-year-old girl who was referred to psychosocial services due to social and psychological difficulties, along with significant impairment in face recognition. McConachie concluded that there was no evidence of a brain lesion or physical damage that would cause AB's inability to identify faces.

Acquired prosopagnosia has two sub-types: apperceptive prosopagnosia and associative/amnestic prosopagnosia (Davies-Thompson et al., 2014). Individuals with apperceptive prosopagnosia lack the ability to perceive a facial configuration and therefore are unable to identify the face. On the other hand, individuals with associative/amnestic prosopagnosia can perceive a face but are unable to make any meaning of the face. The first case of acquired prosopagnosia was documented in 1867 by Quaglino & Borelli (Della & Young, 2003), although the subtype of acquired prosopagnosia is unclear. Mr. LL, the 54-year-old patient, suffered a right hemisphere stroke from a fall. After recovery from unconsciousness, Mr. LL was blind in both eyes but over time, refined his ability to see. However, he showed color blindness and the inability to recognize buildings in his town, along with faces.

Prosopagnosia symptoms are generally contained to an inability to recognize other people, especially close friends and family, along with a difficulty in retaining new faces. Additionally, some people with prosopagnosia have a hard time recognizing their own self in a mirror. Not remembering a face or the incapacity to describe faces makes it difficult for those with prosopagnosia to create relationships and pursue a normative social life. So, to identify others, they are likely to enable non-facial cues, including speech and shared experiences, or utilize object recognition since that neural system is unaffected by prosopagnosia. But, despite that, social interaction is still heavily affected by prosopagnosia, often resulting in poor mental health.

Prosopagnosia in the brain

The face is the first encounter with someone's identity and, thus, numerous brain areas are allocated for face processing. The core face-selective areas lie in the occipitotemporal cortex and consist of the fusiform face area (FFA), the posterior superior temporal sulcus (pSTS), and the occipital face area (OFA)

(Nikel et al., 2002). In prosopagnosia, although the core face-selective areas are located in both hemispheres, the damaged brain areas are mainly in the right hemisphere, specifically the fusiform gyrus which can be found at the lower surface of the occipital and temporal lobes. The fusiform gyrus plays a large role in visual processing, with the FFA within it being strongly associated with facial identity processing. The pSTS lies adjacent to the FFA and plays a role in processing facial expressions, while the OFA processes the components of the face and facial configuration, such as the eyes and mouth. Nevertheless, patients with acquired prosopagnosia often have lesions in both hemispheres even though just a lesion in the right hemisphere is enough to cause prosopagnosia (Landis et al., 1986).

Diagnosis and presentation of prosopagnosia

A prosopagnosia diagnosis is given if someone performs poorly on face recognition tests, such as the Cambridge Face Memory Test (CFMT) which measures memory for unfamiliar face identities. The CFMT shows participants six target faces, and then participants are asked to choose between three faces in a forced-choice task, where one of the faces is a target face. The CFMT capitalizes on the strengths of the Benton Facial Recognition Test (BFRT) and the Recognition Memory Test for Faces (RMF). In the BFRT, participants are presented with a face, then have to determine which three faces, out of six faces in differing orientations, match the target face. However, evidence found by Duchaine & Nakayama (2006) shows that individuals with prosopagnosia were able to perform normally on the BFRT. In the RMF, participants are shown 50 target images briefly, then are presented with 50 forced-choice items that include a target face and a distractor face. However, since non-facial feature differences are included in the images, prosopagnosia individuals have also been shown to score similarly to the controls on this test (Duchaine & Nakayama, 2006).

Additionally, prosopagnosia can also be assessed through self-report measures, although this was not preferred due to inaccuracies and bias. Neuroimaging techniques, such as functional and structural brain imaging, can lend insight into the activity of the face-processing areas of the brain. For example, a major face processing indicator that can be found from an EEG is the ERP component N170 (Susilo & Duchaine, 2013). In normally functioning patients, the N170 shows a larger amplitude response to faces than nonfaces. For some prosopagnosia patients, the N170 shows the same response for both faces and nonfaces, demonstrating the lack of face selectivity. It is to be noted that the differences in N170 responses differed per patient, likely due to the heterogeneity of prosopagnosia, so the imaging technique cannot be relied on to provide a diagnosis. A prosopagnosia diagnosis cannot be determined by imaging alone as the relationship between the neural and cognitive mechanisms remains uncertain.

Prosopagnosia symptoms predominantly manifest in the inability to recognize familiar faces but often affect other cognitive abilities such as a reduction in the face inversion effect (Busigny & Rossion, 2010). The face inversion effect refers to the difficulty in recognizing faces when they are upside down, and since individuals with prosopagnosia already have difficulty in recognizing faces in general, the face inversion effect is diminished. Prosopagnosia also often co-occurs with memory impairments, visual processing difficulties, and social hindrances. With social interactions being the crux of society, those with prosopagnosia often find themselves struggling with social relationships and everyday life, sometimes leading to social anxiety disorder (Yardley et al., 2008). Additionally, a DP case previously examined by Jones & Tranel (2001) expressed difficulties in drawing and topographical orientation.

Updates in prosopagnosia research and treatment

After years of difficulty in diagnosing prosopagnosia, many neuroscientists have suggested new approaches to diagnosing and researching the disorder. Since the CFMT misses around 50-65% of cases when identifying individuals with prosopagnosia, Burns et. al (2022) proposed using the prosopagnosia index as the main method for determining a diagnosis as it has offered a correct diagnosis in almost all cases of prosopagnosia. The 20-item prosopagnosia index (PI20), otherwise known as a prosopagnosia symptom questionnaire, is a self-report assessment that asks participants to indicate how strongly they agree that they experience prosopagnosia symptoms in their daily lives.

Although there is no cure or definitive treatment for prosopagnosia, numerous possibilities have been explored and among them, training in perception and face naming has resulted in the improvement of a patient's prosopagnosia symptoms. Brunsdon et. al (2006) conducted a treatment case for an 8-year-old DP patient, AL. AL possessed a range of cognitive and visual recognition impairments, such as the inability to recognize photos of family members, errors in picture-naming tasks, and below-average visuospatial ability. Thus, the aim of AL's treatment was to improve his ability to perceive and discriminate facial features, decrease his reliance on non-facial cues, and improve his ability to recognize the faces of family members. Treatment involved face naming with photograph stimuli that varied in facial features with an analysis of facial features over the course of a few weeks, and the results revealed improvements in AL's face recognition.

In cases of acquired prosopagnosia, which can be brought on by a right-hemisphere stroke, published cases have reported that lesions in face-processing areas are resistant to treatment, but some recovery is possible through rehabilitation (Cousins, 2013). Even so, the rehabilitation treatment focuses on supporting the patient's quality of life and training the use of nonvisual cues when it comes to face recognition, rather than restoring original cognitive and recognition abilities.

Conclusion

The incapacity to identify familiar faces caused by prosopagnosia is detrimental to an individual's social life, relationships, and everyday functioning. The disorder predominantly affects the face processing areas in the right hemisphere of the brain, including the fusiform face area, the posterior superior temporal sulcus, and the occipital face area. But, prosopagnosia doesn't only arise from brain damage or lesions — it can also be genetic and appear without any physical deterioration of the brain. Diagnosis has been challenging due to the lack of documentation and knowledge correlating the cognitive and neural mechanisms of prosopagnosia, but face recognition tests, neuroimaging techniques, and self-report tests have allowed neuroscientists to piece together diagnoses. Although there is no possible preventative measure, yet, previous and current research has made significant progress in identifying prosopagnosia as soon as possible, and therefore a prompt start to treatment. Once more, there is no cure for prosopagnosia, but perception training has shown promising results in improving face recognition and rehabilitation aims to get prosopagnosia patients' quality of life back to before through nonvisual-cue training. Further research is necessary to better understand the underlying mechanisms of prosopagnosia, develop more precise diagnostic tools, and explore interventions to improve the lives of those with prosopagnosia. By addressing these matters, there is a potential for individuals with prosopagnosia to experience progress in their social lives and mental well-being.

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